EXPERIMENT 2

Semantic Search using Cosine Similarity

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AIM

To develop a semantic search system by comparing document vectors and a query vector using cosine similarity.

PROCEDURE

1. Import TfidfVectorizer and cosine_similarity.
2. Create sample documents and define a search query.
3. Convert the documents and query into TF-IDF vectors.
4. Calculate cosine similarity between the query and document vectors.
5. Display the similarity scores and the best matching document.

PROGRAM

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "The sun gives light and heat.",
    "Python is used for programming.",
    "A computer can process information."
]

query = "Which language is used to write programs?"

vectorizer = TfidfVectorizer()
doc_vectors = vectorizer.fit_transform(documents)
query_vector = vectorizer.transform([query])

similarity = cosine_similarity(query_vector, doc_vectors)[0]

print("Query:", query)
print("\nDocument Similarity:")
for i, score in enumerate(similarity):
    print(documents[i], "->", round(score, 3))

best = similarity.argmax()
print("\nBest Match:")
print(documents[best]
```

OUTPUT

```
print("Search Query:", query)
print("\nCosine Similarity Scores:")

for i, score in enumerate(scores[0]):
    print(documents[i], ":", round(score, 3))

best_index = scores.argmax()

print("\nBest Matching Document:")
print(documents[best_index])
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.13/dist-packages (from
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from s
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (fr
Search Query: What is artificial intelligence?

Cosine Similarity Scores:
The sun gives light and heat : 0.0
Python is easy to learn : 0.258
Artificial intelligence makes machines intelligent : 0.516
The moon revolves around the earth : 0.0

Best Matching Document:
Artificial intelligence makes machines intelligent
```

RESULT

Thus, document search was performed by comparing query and document vectors using cosine similarity.

Note: *Install: pip install scikit-learn. TF-IDF is used as a simple local vectorization method.*